

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0517

Parkin 138 kV
April 24, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0517. The Interconnection Customer proposes a 734.4 MW natural gas combined cycle facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on September 11, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 233 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$105,147,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Central Tie / Wythe 138.0 kV
Contingency: Candler's Mountain / Parkin 138.0 kV
Loading: 242.8% (Rating: 286 MVA)
DFAX: 100.0%
Pre-project loading: 14.3%
2. Facility: Amargo / Brookville 138.0 kV Line
Contingency: Parkin / Opossum Crk 138.0 kV
Loading: 238.7% (Rating: 272 MVA)
DFAX: 100.0%
Pre-project loading: 31.3%
3. Facility: Central Tie Substation / Opossum Crk 138.0 kV
Contingency: Parkin / Candler's Mtn 138.0 kV
Loading: 236.8% (Rating: 297 MVA)
DFAX: 100.0%
Pre-project loading: 10.3%
4. Facility: Parkin / Opossum Creek 138.0 kV
Contingency: Parkin / Candler's Mtn 138.0 kV
Loading: 231.4% (Rating: 308 MVA)
DFAX: 100.0%
Pre-project loading: 6.7%
5. Facility: Bonsboro / Amargo 138.0 kV
Contingency: Parkin / Opossum Creek 138.0 kV
Loading: 223.3% (Rating: 281 MVA)
DFAX: 100.0%
Pre-project loading: 37.6%

6. Facility: Brookville / South Lynchburg 138.0 kV
Contingency: Parkin Substation / Opossum Creek 138.0 kV
Loading: 220.2% (Rating: 310 MVA)
DFAX: 100.0%
Pre-project loading: 16.6%
7. Facility: Candlers Mountain / South Lynchburg 138.0 kV
Contingency: Parkin / Opossum Crk 138.0 kV line
Loading: 215.4% (Rating: 322 MVA)
DFAX: 100.0%
Pre-project loading: 12.9%
8. Facility: Parkin / Canlers Mtn 138.0 kV
Contingency: Parkin / Opossum Creek 138.0 kV
Loading: 206.7% (Rating: 345 MVA)
DFAX: 100.0%
Pre-project loading: 6.0%
9. Facility: South Creek / Wythe 138.0 kV
Contingency: Parkin / Candlers Mountain 138.0 kV
Loading: 140.0% (Rating: 321 MVA)
DFAX: 46.1%
Pre-project loading: 34.6%
10. Facility: Boonsboro / Nacoochee 138.0 kV L
Contingency: Opossum Crk / Parkin 138.0 kV Line
Loading: 127.4% (Rating: 379 MVA)
DFAX: 51.7%
Pre-project loading: 27.1%
11. Facility: Nacoochee / STexa 138.0 kV
Contingency: Opossum Crk / Parkin 138.0 kV
Loading: 126.8% (Rating: 382 MVA)
DFAX: 51.7%
Pre-project loading: 27.4%
12. Facility: South Colton / Flowood 138.0 kV Line
Contingency: Parkin / Candlers Mtn Substation 138.0 kV
Loading: 118.3% (Rating: 178 MVA)
DFAX: 40.6%
Pre-project loading: 49.1%
13. Facility: S Crk / Intel 138.0 kV L
Contingency: South Crk / Aibnito Pds 138.0 kV Line
Loading: 116.9% (Rating: 253 MVA)
DFAX: 17.0%
Pre-project loading: 67.6%
14. Facility: Shenandoah / Waworth 138.0 kV
Contingency: Bear / Clover 345.0 kV line
Loading: 111.7% (Rating: 231 MVA)
DFAX: 7.5%
Pre-project loading: 87.8%
15. Facility: Aibonito ds Substation / Intel 138.0 kV
Contingency: South Creek / Aibonito Pds 138.0 kV
Loading: 111.6% (Rating: 216 MVA)
DFAX: 8.2%
Pre-project loading: 83.7%
16. Facility: Glenannan Substation / Cal Sierra Development 345.0 kV
Contingency: Glenannan Substation / Roberts Sw Sta 345.0 kV

- Loading: 111.3% (Rating: 1,319 MVA)
DFAX: 7.1%
Pre-project loading: 107.4%
17. Facility: Bixby Hill / Tombstone 500.0 kV
Contingency: Sharyland / Tombstone 345.0 kV
Loading: 106.3% (Rating: 3,937 MVA)
DFAX: 8.0%
Pre-project loading: 104.8%
18. Facility: McCartney / Gaston 345.0 kV
Contingency: Roanoke apids / McCartney 345.0 kV
Loading: 105.1% (Rating: 1,322 MVA)
DFAX: 6.2%
Pre-project loading: 101.7%
19. Facility: Tangy Substation / Ceiba Baja 7012 138.0 kV
Contingency: Clover Substation / Bear 345.0 kV
Loading: 104.8% (Rating: 190 MVA)
DFAX: 6.5%
Pre-project loading: 79.9%
20. Facility: Carrabelle / Louisa Generation Facility 345.0 kV
Contingency: Louisa Generation Facility / North Anna 345.0 kV Line
Loading: 103.6% (Rating: 1,907 MVA)
DFAX: 11.8%
Pre-project loading: 99.1%
21. Facility: South Texa / Frnklin-Elk Grove 138.0 kV
Contingency: Desert Outpost / S Texa 138.0 kV Line
Loading: 103.0% (Rating: 246 MVA)
DFAX: 15.6%
Pre-project loading: 56.4%
22. Facility: Shingletown / Riva Road Substation 345.0 kV L
Contingency: Taber / Glenannan 345.0 kV
Loading: 103.0% (Rating: 1,628 MVA)
DFAX: 6.2%
Pre-project loading: 100.2%
23. Facility: Nauvoo / Roberts Sw Sta 345.0 kV
Contingency: Nauvoo / City West Diesel Plant 345.0 kV
Loading: 102.9% (Rating: 1,505 MVA)
DFAX: 7.4%
Pre-project loading: 99.3%
24. Facility: Glenannan / Roberts Sw Sta 345.0 kV Line
Contingency: Glenannan / Cal Sierra Development 345.0 kV
Loading: 102.0% (Rating: 1,461 MVA)
DFAX: 7.7%
Pre-project loading: 98.1%
25. Facility: Desert Outpost / Monel 138.0 kV L
Contingency: Rustburg / Brush Tavern 138.0 kV
Loading: 100.8% (Rating: 327 MVA)
DFAX: 9.6%
Pre-project loading: 79.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Central Tie / Wythe 138.0 kV	Rebuild line with higher-capacity conductor	\$2,315,000	100% (\$2,315,000)
Brookville / Amargo 138.0 kV	Rebuild line with higher-capacity conductor	\$736,000	100% (\$736,000)
Central Tie / Opossum Creek Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,176,000	100% (\$2,176,000)
Opossum Creek / Parkin 138.0 kV	Rebuild line with higher-capacity conductor	\$2,135,000	100% (\$2,135,000)
Boonsboro / Amargo 138.0 kV	Rebuild line with higher-capacity conductor	\$5,305,000	100% (\$5,305,000)
Brookille / S Lynchburg 138.0 kV	Rebuild line with higher-capacity conductor	\$2,576,000	100% (\$2,576,000)
Candlers Mountain / S Lynchburg 138.0 kV	Rebuild line with higher-capacity conductor	\$4,649,000	100% (\$4,649,000)
Parkin / Canlers Mtn 138.0 kV	Rebuild line with higher-capacity conductor	\$1,321,000	100% (\$1,321,000)
Wythe / South Creek 138.0 kV Line	Rebuild line with higher-capacity conductor	\$15,276,000	89% (\$13,536,000)
Boonsboro / Nacoochee 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,406,000	76% (\$4,886,000)
Nacoochee / STexa 138.0 kV	Rebuild line with higher-capacity conductor	\$4,968,000	76% (\$3,790,000)
S Colton Substation / Flowood 138.0 kV	Rebuild line with higher-capacity conductor	\$3,861,000	80% (\$3,073,000)
S Crk / Intel 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,376,000	61% (\$3,876,000)
Shenandoah / Waworth 138.0 kV	Rebuild line with higher-capacity conductor	\$27,175,000	74% (\$20,143,000)
Intel / Aibonito Pds 138.0 kV	Rebuild line with higher-capacity conductor	\$22,273,000	70% (\$15,691,000)
Cal Sirra Development / Glenannan 345.0 kV	Rebuild line with higher-capacity conductor	\$23,366,000	6% (\$1,475,000)
Bixby Hill / Tombstone 500.0 kV	Reconductor line	\$68,962,000	6% (\$4,352,000)
Mccartney / Gaston 345.0 kV	Reconductor line	\$21,239,000	5% (\$992,000)
Tangy / Ceiba Baja 7012 138.0 kV	Reconductor line	\$2,346,000	66% (\$1,554,000)
Louisa Generation Facility / Carrabelle 345.0 kV	Reconductor line	\$40,198,000	8% (\$3,074,000)
South Texa / Frnklin-Elk Grove 138.0 kV	Reconductor line	\$5,119,000	59% (\$3,009,000)
Riva Rd / Shingletown Substation 345.0 kV	Reconductor line	\$28,395,000	6% (\$1,838,000)
Nauvoo / Roberts Sw Sta 345.0 kV	Reconductor line	\$13,456,000	7% (\$902,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Glenannan / Roberts Sw Sta 345.0 kV	Reconductor line	\$19,110,000	6% (\$1,198,000)
Desert Outpost / Monel 138.0 kV L	Reconductor line	\$896,000	61% (\$545,000)
	Total allocated to project		\$105,147,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.